A Physio-Adaptive System for Real-Time Flow Detection in Competitive Esports

Master thesis

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Delivery date	02.02.2027
Semester of study	Winter semester 2027, 9th Semester

University of Duisburg-Essen

Chair of Information Systems and Artificial Intelligence in Marketing

Faculty of Computer Science

Abstract (only for Bachelor/Master thesis)

The abstract serves to give the reader a rough overview of the content (brief problem definition, solution approaches and possibly the key findings).

- The abstract should be a **maximum of half a page** in length.
- The abstract should always be written in English.
- This section must be deleted for seminar papers.

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In a seminar thesis, a list of figures should only be included if **more than three figures** are included.

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In a seminar thesis, a list of tables should only be included if **more than three tables** are included.

List of Abbreviations
(only for Bachelor/Master thesis)

BPM	Business Process Management
RPA	Robotic Process Automation

A frameless table has been inserted for the list of abbreviations. Once the thesis has been completed, it can be sorted in ascending order according to the first column without having to pay attention to the correct order during creation.

If the list of abbreviations remains empty, this section can be deleted.

1 Introduction

Electronic sports (e-sports) have grown into a major global domain, with industry reports describing a worldwide audience in the hundreds of millions and a market value above USD 2 billion (Newzoo, 2024). In this highly competitive context, the psychological state of *flow* is considered a key performance factor. Within flow theory, flow emerges when perceived player skill and task challenge are well balanced (Csikszentmihalyi, 1975; Moneta, 2012). This thesis focuses on *League of Legends* (LoL) as its specific e-sports context because LoL combines a large competitive ecosystem with clearly structured ranked tiers, team-based coordination, rapid decision-making, and frequent high-pressure events such as fights and objective contests (Gong et al., 2020; Rijnders et al., 2022; Riot Games, 2009). These characteristics make LoL a suitable case for examining how flow can be detected and protected in a demanding real-time game environment.

From a sensing perspective, the thesis places particular emphasis on electrocardiography (ECG). ECG-derived heart rate and heart rate variability (HRV) features provide a comparatively unobtrusive way to capture autonomic arousal and cognitive load during gameplay, and prior work has demonstrated that flow-related states can be classified from HRV-based physiological signals (Rissler et al., 2023). Because ECG can be recorded with a chest belt without requiring players to interrupt play or wear highly intrusive equipment, it is especially relevant for e-sports settings in which comfort, movement freedom, and competitive integrity are important. In addition to ECG, webcam-based facial recordings provide non-invasive affect-related cues that can be synchronized with gameplay and physiological data in gaming contexts (Kutt et al., 2022). This combination motivates a practical sensing setup that links ECG-derived HRV features with webcam-based affect features for later live evaluation. This technical foundation supports the development of *physio-adaptive systems* that use real-time physiological and affect-related data to infer user states and trigger digital environmental adaptations (Adam et al., 2024; Irshad et al., 2023).

Despite the demonstrated feasibility of physiological state detection, research regarding specialized interventions in the e-sports domain remains remarkably scarce. As shown by Adam et al., 2024, prior work has explored flow support through live biofeedback to enhance user awareness and emotion regulation during task execution. Likewise, Rissler et al., 2023 demonstrated that binary flow states (high vs. low) can be classified from HRV using machine learning models. However, the unique demands of competitive LoL gameplay—characterized by extreme physiological arousal, rapid-fire decision-making, team communication, and specialized motor patterns—are not yet sufficiently addressed in these enterprise-focused approaches (Bányai et al., 2019; Poulus et al., 2021; Rijnders et al., 2022).

To address this gap, this thesis investigates the design requirements and technical implementation of a physio-adaptive system specifically tailored to protect and sustain flow in e-sports. By moving beyond general classification models, this study seeks to identify specialized, context-aware mechanisms that effectively assist e-sports athletes in maintaining peak performance. Accordingly, the study addresses the following research questions:

- **RQ1:** How can ECG and webcam-based data be utilized to accurately distinguish between high and low flow states in a competitive LoL environment?
- **RQ2:** What are the specific design requirements and intervention mechanisms for a physio-adaptive system that effectively assists LoL players in maintaining their flow state?

To answer these questions, the thesis follows a multi-phase mixed-methods design. First, a comprehensive baseline of system requirements is derived from the literature and a series of semi-structured interviews triangulated across three distinct cohorts: active competitive League of Legends (LoL) players, game design experts, and physiological computing specialists. These diverse empirical perspectives serve to identify context-aware adaptive measures for the e-sports environment, directly informing the development of a functional working prototype. During model development, the classification pipeline draws on multimodal training data and established physiological flow-classification frameworks (Irshad et al., 2023; Rissler et al., 2023). In the final live evaluation, however, the prototype is restricted to the signals available from the deployed setup—namely ECG-derived features and webcam-based features only. Finally, the prototype's practical utility and the effectiveness of its adaptive interventions are evaluated through follow-up qualitative interviews and functional testing. Throughout the study, qualitative data are analyzed using Grounded Theory Methodology (GTM) (Recker, 2021) to anchor technical design decisions in the lived experiences and competitive requirements of the athletes (Delve & Limpaecher, 2024).

2 Foundations

Directly below each main section, there is space for a brief introduction to the content.

In the foundation's section, important terms, concepts, frameworks, etc., are introduced and defined.

2.1 Outline

The heading structure is hierarchical. The section headings are defined in the style sheets by name as heading 1, heading 2, etc.

Please do **not** use a fourth-level heading.

2.2 Figures and Tables

Figures are **always** inserted centered, with a name and, if applicable, a source.



Figure 1: Figure name (Source if applicable)

It is important to always **explain** figures in the text and to refer to the figures in the text (see Figure 1). This also applies to tables. In the case of third-party figures and tables, the original source must also be indicated (e.g., in brackets at the end of the title). Tables should also be centered and labeled (see Table 1).

Please ensure that the font, color and style of figures and tables are **consistent**.

Table 1: Table name (Source if applicable)

Format template for listings and numbering with multiple levels:

- First level
  - Second level
    - Third level

2.3 Content

Your text should be divided into paragraphs using figures and tables. It should be noted that a paragraph captures a thematic idea, whereby the core statement is usually found at the beginning of the paragraph and, from this, is structured by further discussions within the paragraph.

3 Related Work

This section summarizes the results of other research related to your work. In addition to the textual description, please also use tables for this purpose.

Note: The identified research results should be discussed in the discussion section with the results of the thesis.

The Chair of Information Systems and Artificial Intelligence in Marketing uses the **7th edition of the citation style of the American Psychological Association (APA)**.

In order to give you an insight into how to cite appropriately in your thesis, the following lines serve as an orientation guideline for the citation system. Adhering to the subse-quent citation rules is vital to crafting a successful and adequate paper. For clarification purposes, several examples are shown in the further procedure.

3.1 Indirect Citations at the end of a sentence

If you write a statement or argument in your thesis which is based on the idea or result from another author, you will have to state the author at the end of the sentence. The style of a reference to other scientific work is often not dependent on the type of publication but rather determined by the number of authors.

3.1.1 Single author

Theory of mind enables the prediction of others' behavior beyond familiar patterns, especially in cases involving unfamiliar or mistaken actions (**liszkowski_using_2013**).

3.1.2 Two authors

Support for the universality thesis requires that facial expressions of basic emotions are consistently recognized across cultures with high accuracy rates (**nelson_universality_2013**).

3.1.3 More than two authors

Young children's early prosocial behavior may stem from their innate interest in others, early imitative tendencies, or emerging capacity to understand others' emotional states (**svetlova_toddlers_2010**).

3.2 (Indirect) Inline citation style

If you highlight a result or an idea of some other authors, you can also use inline citations. Make sure that the reference is positioned close to the actual idea of the author(s).

3.2.1 Single author

izard_many_2010<empty citation> emphasized that although emotion research has flourished in recent decades, there is still no commonly accepted definition of emotion.

3.2.2 Two authors

telle_positive_2016<empty citation> argued that positive empathy also contributes to prosocial behavior, a topic that has received relatively little research attention.

3.2.3 More than two authors

brownell_peers_2002<empty citation> found that from around age two, children begin to engage in cooperative play that relies on their developing ability to understand and respond to the mental states of peers.

3.3 Direct citations

In addition, for direct citations for which you cite a complete thought you have to indicate this by using quotation marks. For those cases, you also have to specify page numbers. These rules must be applied no matter what you cite (e.g., journals, conferences, books). The style for direct citations is demonstrated by the following examples:

3.3.1 Direct citation of an entire sentence

"[...] children during the second and third years of life do participate in the work of adults." (**rheingold_little_1982**).

Note: With "[…]" you highlight that you left out information originally contained in the citation but less relevant for your thesis. The meaning must not be changed by using this technique.

3.3.2 Direct citation of single information in a sentence

Already in the earliest stages of development, infants engage in structured interactions with others, often organized as "coordinated, turn-taking sequences" (**warneken_cooperative_2006**).

4 Research Method

4.1 Section 4.1

4.1.1 Section 4.1.1

For Bachelor's and Master's theses, a subdivision up to the third level makes sense. A fourth level (e.g., 4.1.1.1) must be omitted in order to maintain the clarity of the structure. Use subheadings instead.

Subheading

4.1.2 Section 4.1.2

Important: If you also structure 4.1.1 under 4.1, a subheading 4.1.2 is automatically required, as a subheading alone is not permitted.

4.2 Section 4.2

4.2.1 Section 4.2.1

4.2.2 Section 4.2.2

5 Results

It is of major importance to present the results of your research in detail. Therefore, this section should be the main part of your thesis.

6 Discussion

Your results must not only be presented, but also discussed. Emphasize the (theoretical and practical) impact of your work on the research area in question (also in relation to the results mentioned in the related work section). At the end, do not forget to discuss the opportunities for future work and the limitations of your work.

6.1 Theoretical Implications

6.2 Practical Implications

6.3 Future Work

6.4 Limitations

7 Conclusion

The conclusion is where you wrap up your ideas and leave the reader with a strong final impression. It has several key goals:

- Restate the problem statement addressed in the thesis
- Summarize your overall arguments or findings
- Suggest the key takeaways from your paper

The conclusion should be compared with the abstract (if applicable) and the introduction. Similar to the introduction, the conclusion should make up about 5% of the entire thesis.

Please do not forget to **check the spelling** of the entire thesis. You need to write your thesis in **English**. Please use ”**American English**”.

References

Please use **APA 7th edition** as citation style for your thesis.

General information can be found here: https://apastyle.apa.org/

In the following, you will find a sample reference list as a guide:

Adam, M. T. P., Bonenberger, L., Gimpel, H., & Lanzl, J. (2024). Human-centered design and evaluation of a neurois tool for flow support. *Journal of the Association for Information Systems*, *25*(4), 936–961. https://eref.uni-bayreuth.de/id/eprint/88160/

Bányai, F., Griffiths, M. D., Király, O., & Demetrovics, Z. (2019). The psychology of esports: A systematic literature review. *Journal of Gambling Studies*, *35*(2), 351–365. https://doi.org/10.1007/s10899-018-9763-1

Csikszentmihalyi, M. (1975, January). Flow: The psychology of optimal experience.

Delve, L., Ho, & Limpaecher, A. (2024, October). *Open, axial, and selective coding in qualitative research: A practical guide*. Practical Guide to Grounded Theory. Retrieved February 7, 2026, from https://delvetool.com/blog/openaxialselective

Gong, L., et al. (2020). League of legends: A study on the ranking and matchmaking system. *Proceedings of the 2020 International Conference on Computing and Data Science (CDS)*, 245–250. https://doi.org/10.1109/CDS49703.2020.00056

Irshad, M. T., Li, F., Nisar, M. A., Huang, X., Buss, M., Kloep, L., Peifer, C., Kozusznik, B., Pollak, A., Pyszka, A., Flak, O., & Grzegorzek, M. (2023). Wearable-based human flow experience recognition enhanced by transfer learning methods using emotion data. *Computers in Biology and Medicine*, *166*, 107489. https://doi.org/10.1016/j.compbiomed.2023.107489

Kutt, K., Drążyk, D., Żuchowska, L., Szelążek, M., Bobek, S., & Nalepa, G. J. (2022). BIRAFFE2, a multimodal dataset for emotion-based personalization in rich affective game environments. *Scientific Data*, *9*(1), 279.

Moneta, G. B. (2012). On the measurement and conceptualization of flow. In S. Engeser (Ed.), *Advances in flow research* (pp. 23–50). Springer New York. https://doi.org/10.1007/978-1-4614-2359-1_2

Newzoo. (2024). *Global esports & live streaming market report 2024* (tech. rep.). Newzoo. https://newzoo.com/resources/blog/esports-market-trends-2024

Poulus, D., Coulter, T., Trotter, M., & Polman, R. (2021, October). A qualitative analysis of the perceived determinants of success in professional esports athletes [Publication details to be added (e.g., journal/conference, volume/issue, pages, DOI/URL).].

Recker, J. (2021). Research methods. In *Scientific research in information systems: A beginner's guide* (pp. 87–160). Springer. https://doi.org/10.1007/978-3-030-85436-2_5

Rijnders, F., Wallner, G., & Bernhaupt, R. (2022). Live feedback for training through real-time data visualizations: A study with league of legends. *Proc. ACM Hum.-Comput. Interact.*, *6*(CHI PLAY). https://doi.org/10.1145/3549506

Riot Games. (2009). League of Legends [Version 2026].

Rissler, R., Nadj, M., Li, M. X., Loewe, N., Knierim, M. T., & Maedche, A. (2023). To be or not to be in flow at work: Physiological classification of flow using machine learning. *IEEE Transactions on Affective Computing*, *14*(1), 463–474. https://doi.org/10.1109/TAFFC.2020.3045269

List of Permitted Aids and Documentation

Obligations by the supervisor (first reviewer):

The supervisor (first reviewer) informs the student of the list of permitted aids. The date will be specified in order to be able to prove the exact status of the list beyond doubt. This is the date on which the list was announced to the student. If the list is updated by the supervisor (first reviewer) - e.g., by adding further permitted aids - the date must be updated and the student affected must be informed of the change immediately. If an update is made on the same day, the version of the list can be expressed by an addition (e.g., 02.09.2024b).

List of permitted aids that can be used by the student:

- Microsoft Word or comparable word processing programs **without** AI plug-ins/add-ins (e.g. Copilot)
- Google or comparable search engines
- Grammarly or comparable AI tools for the correction of grammar
    - If the student plans to use comparable tools, these **must** be accepted by the supervisor in each case.
- DeepL or comparable AI tools for the translation of texts
    - If the student plans to use comparable tools, these **must** be accepted by the supervisor in each case.
- DALL-E or comparable image-generating AI tools
    - If the student plans to use comparable tools, these must be accepted by the supervisor in each case.
- ChatGPT and other Large Language Models that can be accessed via the **ChatAI access of the UDE** can be used
- GitHub Copilot
    - If the student plans to use comparable tools, these **must** be accepted by the supervisor in each case.

Obligations by the student:

Any use of permitted aids must be **indicated**. If the same selection of tools has been used for all chapters, this must be indicated on the first page of the introduction by a footnote reference. If the student wishes to indicate the use of permitted aids for different purposes in different sections, this must be indicated by a footnote reference on the first level of the heading.

Prompts and chat answers / results for LLMs must be submitted separately. Submit separate files for different models and only one file per model. This one file has to contain the entire chat, including all answers and results for this specific source. In the research method section, a **separate subsection** must also describe which permitted aids were used and how

they were used. The following **sample documentation** can serve as a guide:

Section	Tool (possibly with technical restric-tions)	Purpose of use in the context of the exami-nation perfor-mance or text to be submitted	Proof or labeling obligation	Exemplary Prompt	Exemplary Remarks
All sections	Grammarly	Correction of grammar	Identification of the text passages concerned in accordance with the general citation rules	-	Accepted the grammatical corrections in accordance with the general citation rules
All sections	DeepL	Translation of text passages	Identification of the text passages concerned in accordance with the general citation rules	-	Relied on the translated text in accordance with the general citation rules
Section 1 - Intro-duction	ChatGPT and other Large Language Models that can be accessed via the ChatAI access of the UDE can be used	Creation of a practical challenges list	Identification of the text passages concerned in accordance with the general citation rules; prompts and AI-generated answers as an appendix	"Name the top 5 practical challenges in the field …" *Answer:* …	Two practical challenge suggestions were used after review based on own reading and understanding of the papers, remaining challenges from other sources

Section 2.3 - Related Work on …	ChatGPT and other Large Language Models that can be accessed via the ChatAI access of the UDE can be used	Summary creation of related work	Identification of the text passages concerned in accordance with the general citation rules; prompts and AI-generated answers as an appendix	"Summarize the papers A, B, C, and D in 6 sentences …" *Answer:* …	Initial summary created but reworked based on own reading and understanding of the papers
Figure 3	ChatGPT and other Large Language Models that can be accessed via the ChatAI access of the UDE can be used	Incorpo-ration of figure in edited or unedited form into the text	Identification of the tables, graphics and verbaliza-tions; Supplemen-tary indication of the prompts used; prompts and AI-generated answers as appendix	"Create a figure that …" *Answer:* …	1:1 usage of the created figure in accordance with own reading and understanding of the papers
Figure 4	DALL-E or comparable image-generating AI tools	Incorpo-ration of images or graphics in edited or unedited form into the text	Identification of the relevant rep-resentation or graphic in accordance with the general citation rules; prompts and image data generated by the AI in the appendix	"Create a graphic that …" *Answer:* …	1:1 usage of the created graphic in accordance with own reading and understanding of the papers

Section 5 - Results	ChatGPT and other Large Language Models that can be accessed via the ChatAI access of the UDE can be used	Data Analysis	Written, descriptive documentation of the use of the tool and submission of the complete entire source code; Supplementary indication of the prompts used; prompts and AI-generated answers as appendix	"Create code for a boxplot visualization …" *Answer:* …	Initial Python code created and leveraged based on own understanding
Code	GitHub Copilot	Code Generation	Written, descriptive documentation of the use of the tool and submission of the complete entire source code;	-	Code suggestions leveraged based on own understanding

Appendix (if applicable)

An appendix is necessary if materials that relate to the thesis as a whole or to larger parts of it are difficult to include in the main body of the thesis. This is particularly the case if they would disrupt the overall coherence of the thesis due to their size. The content of the appendix must not contain anything that is necessary for understanding the text. It should not be used to expand the thesis and make it appear "thicker".

The appendix is suitable for supplementary documents and materials, especially if these are difficult or impossible for the reader to access, e.g., unpublished company documents.

Particularly in the case of empirical work, the appendix can be used to identify the data sets, mathematical-statistical methods or programs used in more detail.

If the appendix consists of several elements, the individual elements must be separated from each other by numbering.

If the appendix remains empty, this page can be deleted.

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